No Public Designation Names a Detection Measurement for the Simulants Chemical Warfare Agent Sensing Is Built On, and Descriptor Comparison Cannot Supply the Missing Validation

Ku Kang[a], Jeongyun Kim[a], Miae Kim[a], Doo-Hee Lee[b], Jin Yoo[a], Yoon Jeong Jang[a,*]

[a] *Chemical, Biological, Radiological and Nuclear (CBRN) Defense Research Institute, Ministry of National Defense, Seoul, Republic of Korea*
[b] *Department of Civil Engineering and Environmental Sciences, Korea Military Academy, Seoul 01805, Republic of Korea*

Abstract

Chemical warfare agent detection is developed almost entirely on simulants: compounds accepted as agent-like and safe to handle. We tested that acceptance. In the public standard-setting record, two of the three incumbent simulants are designated for measurements that are not detection and the third nowhere we could reach; the Army procedure governing simulant selection is not public, so this bounds the claim. For eleven authentic agents we then computed structural and GFN2-xTB electronic descriptors and asked, by exact permutation over every designation, whether those three sit closer to the agents than the panel's other non-agents. On the primary thirty-five-compound panel, which adds five compounds the sensor literature also calls simulants, no agent has a valid match and no test reaches significance. Resampling the descriptors gives the incumbent three a median of zero matched agents against five for a set of eight containing them, and neither count is unusual for its size: the zero sits at $p = 0.48$ against an expected 1.0, the five at $p = 0.07$ against 1.92 once the null contains the incumbents. What the extended panel changes is the comparison rather than the chemistry, since diisopropyl fluorophosphate is nearer to three G agents than any incumbent is. A twenty-eight-compound panel omitting those seven yields four matches and a nominal p of 0.025 that no correction over eighteen tests leaves significant, while relabelling the two pesticides as simulants produces an adjusted 0.0011. A defensible claim must name the measurement it rests on and fix panel, space, statistic and weighting in advance.

Keywords: chemical warfare agent simulant, descriptor space, DMMP, exact permutation test, acute exposure guideline level, dual-use

Environmental implication

Chemical warfare agents are among the most acutely toxic substances in commerce or in storage, and almost everything done to detect, contain or destroy them in the field is developed and qualified on substitutes. Decontamination efficiency is quantified against dimethyl methylphosphonate [1], protective ensembles are qualified on vapour penetration by methyl salicylate [2, 3, 4],

*Corresponding author.
Email address: yjjangchem@korea.kr (Yoon Jeong Jang)

and point detectors are tested against simulants chosen by a procedure that is not public [5, 6]. This work shows that the compounds carrying that weight in the sensing literature hold no public designation naming the measurement they are used for, that one of them has no regulator-derived acute inhalation value at all, and that a descriptor comparison of the kind the field uses to justify such choices does not settle the question either way. The practical consequence is for anyone who reads a simulant result as a statement about the agent: the measurement the compound was validated against, where one exists, is the limit of what that reading supports.

1. Introduction

Work on detecting chemical warfare agents (CWAs) is rarely done with CWAs. Outside a small number of licensed facilities, sensor arrays, sorbents and sampling protocols are developed, optimised and published using simulants: compounds chosen to behave like an agent while being safe enough for an ordinary laboratory. The choice of simulant therefore sets a ceiling on what a published detection result can be taken to mean, and that choice is made by a convention rather than by a measurement.

The dependence is structural, not incidental. The standing reviews of agent destruction and detection organise a literature in which reactivity and sensing chemistry are established on surrogates and extrapolated to agents [7, 8], and the same pattern runs through the chromogenic, fluorogenic and supramolecular recognition chemistry developed for nerve agents [9, 10, 11]. Much field instrumentation descends from a colorimetric sensor-array lineage established on ordinary analytes and odours [12, 13, 14] and transferred to the agent problem, so the compounds an array is characterised on are inherited rather than fixed by the chemistry. Our own work sits on both sides [15, 16, 17], and detection closed against authentic agents is rare enough that surrogate work is where most of the field's evidence is generated.

What a simulant is for differs by subfield, and so does what makes one adequate. For environmental fate and transport the criterion has been reviewed explicitly [18]; in metal–organic framework adsorption and catalytic hydrolysis the simulant establishes the mechanism and the kinetics [19, 20]; in decontamination the organophosphorus simulant dimethyl methylphosphonate (DMMP) is the compound whose removal is quantified [1]. In every case the inference runs from the simulant to the agent, and the warrant for it is a resemblance argument.

The convention is property matching. A recent critical review of liquid, low-toxicity simulants states the criterion plainly: align physicochemical properties such as hydrophobicity, volatility and solubility with those of the agent, with the simulants it assesses described as presenting minimal toxicity risks [21]. That review is confined to liquid-phase sampling, but the same criterion underwrites vapour-phase detection, where the quantity of interest is not partitioning but a chemical response.

Two assumptions are built into this practice, and both are testable. The first is that resemblance in structural and physicochemical descriptors implies resemblance in the property a sensor reports. For adsorption in metal–organic frameworks this has been examined directly: four common simulants and two nerve agents simulated across 2969 frameworks reproduce agent adsorption behaviour only partially, and the agreement depends on which framework and which agent [22]. The analogous question for the electronic properties governing a chemical response has not been asked. A cheminformatic attempt at the selection problem predates this one, in this journal: Lavoie et al. applied the chemical similarity search used in pharmaceutical screening to simulant identification, with the stated aim of finding compounds close to the agents but less toxic [23]. That aim is the one the field still has. What the present work adds is that the compounds the detection

literature settled on were not reached by that procedure or by any comparable one we could find. The second assumption is that a simulant is safe, usually supported by the absence of an agent-class hazard label rather than by a toxicity value, although where simulants are handled at scale the protective equipment is specified against measured permeation data [24]. That is the practice in detection; it is not the practice everywhere. The human-decontamination literature screens candidates against toxicity explicitly and reports the attrition. One systematic review identified 22 simulants and judged 17 unsuitable for human volunteer trials [25]; a later one screened 171 chemicals, discounted 78 on the risk they would pose to volunteers, and assembled the remainder into a property-indexed repository of 39 suitable and 54 provisional compounds [26]. The two literatures select under different criteria because the measurements differ: a decontamination trial is governed by volatility and solubility, a sensor response by molecular recognition.

Both assumptions can be checked against measurements that already exist. The electronic structure of small molecules is accessible at semi-empirical tight-binding cost [27, 28, 29], and acute inhalation toxicity has been derived by a formal regulatory process for several hundred chemicals, including the agents themselves [30]. Ranking compounds along a sparse regulator-derived hazard axis bounded by an explicit applicability domain is the approach we have applied elsewhere [31]; here it is turned on the compounds the detection field uses on itself. We ask where simulants and agents sit relative to one another, how much of that answer is fixed by the data rather than by the analyst's choice of panel, space and statistic, whether the incumbents are safe and whether that claim rests on a value or on an absence, and whether fidelity and safety are coupled.

The first question turned out to have a different shape than we expected. We began by comparing descriptor spaces, found a difference, and then found on audit that it did not survive extending the compound panel, correcting for the number of statistics computed, or removing a descriptor that was an exact function of two others in the same space. What the data support is reported below as such, with the earlier reading alongside it, because the way a nominal result of this kind arises is part of what a reader of this literature needs.

Dual-use statement.. Throughout, the optimisation direction is fixed: fidelity to the detection-relevant properties is to be maximised while phosphorus electrophilicity, acetylcholinesterase reactivity, alkylating capacity and predicted mammalian toxicity are to be minimised. No analysis here ranks compounds by predicted mammalian toxicity. One analysis does order compounds on a reactivity surrogate: Supplementary Section S15 reports the condensed Fukui index at phosphorus for the six phosphorus-bearing compounds of the panel, which is defined with respect to attack on phosphorus and is therefore an acetylcholinesterase-relevant quantity. It covers only compounds whose structures and agent status are already public, it is used to show that the incumbent simulant sits closer to the agents on that axis than the fidelity argument assumes, and it is not extended to any novel structure. The structural filter in Section 2.7 excludes candidates and never orders them. One ordering is published, in Supplementary Section S19, and it ranks compounds that have cleared that filter by resemblance to the agents in the electronic space. Every compound in it is an industrial chemical in commerce that already carries a published regulatory exposure value, which is how it entered the pool at all, and the selection that produced the list is a safety threshold rather than a reactivity one. The alert patterns themselves are the agent-class motifs of the published structural-alert literature cited in Section 2.7, and Section 2.7 states in prose what each tier covers, so releasing the patterns with the code adds no structure-activity information that the text does not already give. We did not extend the toxicity model of Supplementary Section S2 to the organophosphorus class, because a model predicting mammalian lethality for structures not

yet made would invert the direction of the work. That a predictive model trained for safety can be run in reverse is not hypothetical [32], and how much of such a method to disclose is itself a live problem [33]. Mechanistic modelling of organophosphorus lethality is established and separately published, including by the present authors [34]; what is withheld is its use as a design objective over novel structures, not the literature's existence.

2. Methods

2.1. Compound set and the two panels

Eleven authentic agents are common to both panels: sarin GB, soman GD, tabun GA, cyclosarin GF, VX, sulfur mustard HD, lewisite 1 L, hydrogen cyanide AC, cyanogen chloride CK, phosgene CG and chloropicrin PS. Three compounds are treated as the incumbent simulants: DMMP, dichloromethane and dipropylene glycol monomethyl ether (DPGME).

We call these three incumbent rather than designated, and the word is operational rather than evaluative: they are the compounds the chemical warfare agent sensing literature we surveyed runs its arrays and sorbents against, which is a statement about usage and not about standing. That usage is thin for two of them, resting on one metal-oxide sensor study for dichloromethane [35] and on a training patent for DPGME [36], and a different survey could reasonably return a different three. Section 3.1 reports what searching for a designating instrument returned, since that search is a result of this work rather than a premise of it; the panel construction below uses its outcome.

The two panels differ only in the non-agent pool, and the difference is the point. The twenty-eight-compound panel adds fourteen interferents and solvents routinely reported in sensor-array selectivity panels. The thirty-five-compound panel adds, to those, five further compounds used as simulants in the sensor, sorbent and decontamination literature (diisopropyl fluorophosphate DFP, diethyl chlorophosphate DCP, diethyl cyanophosphonate DCNP[37, 38], 2-chloroethyl ethyl sulfide CEES and methyl salicylate), classified as simulants rather than as interferents. DCNP carries the clearest documentation of the five: it is the compound probe chemistries are developed against when the target is tabun, in a hydrogel sensor built on a self-propagating cascade [37] and in a selective optical scheme for tabun mimics [38]. The panel adds, alongside these five, two organophosphorus pesticides, dichlorvos and malathion. We class the pesticides as interferents because the sensor and decontamination literature we surveyed does not use them to stand in for an agent, and flag the decision as contestable on two counts: malathion appears in the NIOSH document's barrier-breakthrough tables, and both pesticides carry the phosphoryl motif that the structural filter of Section 2.7 treats as agent-class. Section 3.2 reports how sensitive the result is to exactly this kind of decision.

The thirty-five-compound panel is the primary one, and the five it adds are the ones the same documents name: methyl salicylate is the Man-in-Simulant Test simulant and the subject of a published test method [2, 3], DFP is recorded in the NIOSH notes as having been used as a vapour simulant for GB, CEES appears in that document's barrier-breakthrough tables, and all five are in current use in the sensor literature. A comparison asking whether three compounds are close to the agents while excluding the obvious alternatives is not a test of those three. We report the twenty-eight-compound panel throughout with equal numerical detail, because what it shows beside the larger one is the paper's result, and not to support a conclusion. All class labels were read from a single canonical table and never assigned by hand.

Panel, space, statistic and descriptor weighting were not fixed before the analysis was run, and this paper's conclusion is that they should be. We state them here so that the standard we

recommend can be applied to the work recommending it. The thirty-five-compound panel is primary because the five compounds it adds appear in the same documents as the incumbents, and two of them under a stronger grade of standing than dichloromethane has. The merged space is reported alongside both component spaces throughout rather than selected among them. Six distance statistics were computed and none was nominated in advance, which Section 3.2 reports the consequence of and Supplementary Table S2 prices. Equal weight per descriptor was the default we inherited rather than one we chose, and Supplementary Section S12 reports the alternative.

2.2. Descriptors

Structural and physicochemical descriptors (51 after excluding run metadata) were computed with RDKit [39] from canonical SMILES, and include the Crippen atomic-contribution estimates of logP and molar refractivity [40], the fragment-based topological polar surface area [41] and the κ shape indices [42]. Interpretable descriptors were used throughout in preference to a substructural fingerprint such as ECFP [43], because the quantity of interest here is the position of a compound along named physical axes rather than a similarity score.

Ten of the fifty-two are not functions of the SMILES string alone. The RDKit shape descriptors, the three principal moments of inertia and the radius of gyration, asphericity, eccentricity, inertial shape factor, the two normalised principal-moment ratios and the spherocity index, are read from a three-dimensional structure, and nine of them survive the correlation filter into every distance this paper reports. An earlier version of this work took them from the single lowest-energy conformer of an ETKDGv3 ensemble, which made them depend on a choice the paper never declared and which did not reproduce when the descriptor build was repeated. We now take each as a Boltzmann-weighted mean over the whole ensemble at 298 K, with MMFF94 energies or UFF where MMFF has no parameters. For seventeen of the twenty-eight compounds one conformer carries essentially the whole weight and the mean equals the single-conformer value; the difference is confined to the flexible compounds, of which VX and malathion are the extremes. Section S4 reports what the two treatments do to the results, and the deposited `build_descriptors.py -verify` reproduces every numeric column of the descriptor cache from the code. Identifiers, formulae and molecular weights were resolved against PubChem [44]; the EPA CompTox Dashboard [45] was the intended second route but its programmatic interface was not reachable from either network used here, so identities rest on PubChem and on the identifiers printed in the regulatory source documents, and no property value was taken from a prediction service.

Electronic descriptors were computed with GFN2-xTB [28], the second-generation parametrisation of the geometry–frequency–noncovalent tight-binding family [27, 29], in the gas phase; the implicit solvation model available for these methods [46] was not applied, since vapour-phase response is the property of interest. Conformers were generated with ETKDGv3 [47, 48], optimised with MMFF94 [49] (UFF [50] for lewisite 1, the only compound lacking MMFF parameters), then optimised and evaluated at GFN2 level. Ensembles were enumerated from the distance-geometry stage rather than through a dedicated sampling workflow [51], as these descriptors are not strongly conformation dependent. Properties for the 28-compound set were Boltzmann averaged over up to five conformers at 298.15 K with tight optimisation; the larger AEGL screening set used a single conformer with `-opt normal`. On the 28 compounds the two protocols agree to Pearson $r \geq 0.9857$ across the four electronic descriptors, with mean absolute differences of 0.029 to 0.052 eV for the orbital quantities and 0.11 D for the dipole, the largest single dipole discrepancy being 0.95 D at tabun. Which protocol an analysis uses is fixed by

what it compares. The descriptor-space comparison of Section 3.2, which stays inside the 28 compounds, uses the Boltzmann-averaged values. The decoupling analysis of Supplementary Section S19 places the 28 and the AEGL pool in one space, so it uses the single-conformer values for all of them, including the 28; mixing protocols within a distance calculation would put part of the systematic difference above into the distances.

Four electronic descriptors are available for every compound: E_{HOMO}, E_{LUMO}, the global electrophilicity index $\omega = \mu^2/2\eta$ in the definition of Parr et al. [52], and the dipole moment. Four further descriptors exist only for the six phosphorus-bearing compounds: the Mulliken charges at phosphorus and at the phosphoryl oxygen, the Wiberg P=O bond order [53], and the condensed Fukui index f^+(P) for nucleophilic attack at that centre [54, 55, 56]. They are reported in Supplementary Section S15 and excluded from any analysis spanning all 28 compounds.

2.3. Descriptor spaces and distances

Three spaces were compared: structural, electronic and merged.

The electronic space carries three coordinates, not the four that frontier-orbital descriptor sets of this kind usually report. The fourth, the Parr electrophilicity index $\omega = \mu^2/2\eta$ with $\mu = (E_{\mathrm{HOMO}} + E_{\mathrm{LUMO}})/2$ and $\eta = (E_{\mathrm{LUMO}} - E_{\mathrm{HOMO}})/2$ [52], is an exact function of the two orbital energies that accompany it. We confirmed this numerically: across the primary panel $|\omega - \mu^2/2\eta|$ reaches at most 1.3×10^{-5} eV, and 7.1×10^{-15} eV on the twenty-eight-compound panel. The correlation filter described below does not remove it, because $r(E_{\mathrm{LUMO}}, \omega) = -0.899$ falls below the $|r| > 0.95$ threshold; a filter built on linear correlation will pass a variable determined exactly by functional dependence. Three principal components of the standardised four-variable set account for 97.6% of its variance, and removing ω leaves the rank order of pairwise distances almost unchanged (Spearman 0.934 on the primary panel, 0.932 on the smaller one). We therefore take E_{HOMO}, E_{LUMO} and the dipole moment as the electronic space, and report ω and the condensed Fukui index in Supplementary Table S5 without admitting them to any distance. The four reactivity descriptors localised at phosphorus are undefined for the twenty-two panel compounds that contain none, so they cannot enter a panel-wide distance either.

Within each space, descriptors with zero variance were removed, then one of each pair correlated at $|r| > 0.95$ was removed with preference for retaining a protected set (molar refractivity, logP, TPSA, hydrogen-bond acceptor count, four linear-solvation-energy surrogates and the electronic descriptors). The four surrogates are linear rescalings of quantities already in the set and are not Abraham solvation parameters; all four are removed before any distance is taken, three of them by the correlation filter at $|r| = 1.00$ in every space and the fourth by the exact-dependence check on the twenty-eight-compound panel and by correlation at 0.95 on the primary one. No structural descriptor has zero variance. The correlation filter and the exact-dependence check of Section 2.3 together remove 13 of the 51 on the primary panel and 15 on the twenty-eight-compound panel, leaving 38 and 36 structural coordinates respectively; the attrition is therefore not the same on the two panels. Remaining descriptors were standardised and distances taken as Euclidean. Standardisation, and the median nearest-neighbour separation that normalises the distances, are computed over the compounds of the panel in hand. A normalised distance is therefore comparable within a panel and not between panels, which matters wherever the two panels are set side by side; where that comparison is made below it is made on the thirty-five-compound basis. Two shape indices, κ_2 and κ_3, go negative for the smallest molecules (water, $\kappa_3 = -104.0$); repeating every analysis without them returns the permutation values of Section 2.4 to three decimal places (0.026, 0.275 and 0.025), so no result below rests on the pathological indices.

2.4. Nearest-neighbour comparison and permutation test

For each authentic agent we recorded the distance to the nearest incumbent simulant and to the nearest interferent. With fourteen interferents against three simulants, the minimum of fourteen draws is smaller than the minimum of three by construction, so a paired test of the two distances is confounded by group size. We therefore enumerated all ways of designating three of the non-agents as simulants, computed the same statistic for each, and report the rank of the actual designation as an exact p-value. That is $\binom{24}{3} = 2024$ designations on the primary panel and $\binom{17}{3} = 680$ on the twenty-eight-compound panel. The null expectation for "an interferent is closer" follows from the group sizes alone and is 9.63 of 11 on the primary panel and 9.06 on the smaller one.

The hypothesis under test is directional, since property matching is claimed to place simulants closer to the agents than arbitrary compounds would be. The reported p is therefore one-sided, the fraction of designations whose mean d(nearest simulant) − d(nearest interferent) is at least as negative as the actual one's. The choice is consequential and we give all three conventions: scored two-sided as distance from the null mean the same statistic gives 0.035, 0.482 and 0.028 for the structural, electronic and merged spaces, and at twice the smaller tail 0.053, 0.550 and 0.050, so whether the structural value sits inside a conventional threshold depends on a choice a reader may make differently. This is the first of the analyst choices catalogued in Section 3.2 and the only one with a defensible default, the hypothesis as its proponents write it being directional.

2.5. Match criterion

Nearest-neighbour rank alone is not sufficient: in the 40-dimensional merged space of the twenty-eight-compound panel a "nearest" simulant at a distance of 11.8 is not a match. We therefore normalised each distance by the median nearest-neighbour distance of the same space and define a valid match as a normalised distance ≤ 2 together with a distance smaller than that to every compound outside the designated set. The two conditions are not equally fragile: the threshold is a convention, the second condition is not. The criterion is a statement about proximity in this descriptor space and nothing further. It does not certify that a simulant reproduces an agent's response on a given instrument, and a compound that fails it may still suit a measurement governed by one descriptor rather than by their aggregate. On the primary panel the incumbent simulants fail the second condition for all eleven agents, so the count there is zero at every threshold, including an unbounded one. We report the count at 1.5, 1.75, 2.0, 2.5 and 3.0 in Section 3.3, including the one setting at which it changes.

2.6. Descriptor-resampling sensitivity analysis

The conclusions of Section 3.2 turn on differences that are small in absolute terms, 0.12 typical nearest-neighbour separations between DMMP and diisopropyl fluorophosphate at sarin, and a reader is entitled to ask how much of that the descriptor choice carries. We resampled the 42 merged-space descriptors with replacement 1000 times (seed `0xC0DE`), restandardised, recomputed the distance matrix and the median nearest-neighbour separation within each replicate, and recounted the matched agents. Resampling features rather than compounds is the relevant perturbation, because the compound set is fixed by the question and the descriptor set is not. The descriptors are not sampled observations drawn from a defined population, so the replicates are read as a sensitivity analysis over descriptor choice and the reported intervals are percentile ranges across replicates rather than confidence intervals in the sampling sense. The same replicates serve both simulant definitions, so the difference is paired.

Counts obtained from designated sets of different sizes are not directly comparable, because the second condition of the match criterion asks a designated compound to be nearer than every compound outside the designated set, and that condition is easier to meet as the set grows. We therefore computed, for each set size, the distribution of matched agents over designations of that many non-agents drawn from the same panel: exhaustively over all 2024 designations of three, and over 20,000 draws for eight, where 735,471 designations exist. The observed counts are reported against the mean of that distribution and against the proportion of designations reaching at least the observed count.

2.7. Structural filter

A SMARTS filter was written to exclude candidates carrying agent-class motifs, following the structural-alert tradition in which a substructure associated with a mechanism of toxicity serves as a categorical screen [57, 58]; the motifs here are agent-class rather than generic and the filter excludes only. Tier A covers a phosphoryl group bearing a good leaving group (fluoride, cyanide, thiolate or activated aryloxy), arsenicals, sulfur and nitrogen mustard backbones, and cyanide-releasing motifs. Tier B covers phosgene-derived acyl halides and chloroformates, halogenated oximes, α-halo carbonyls, and groups making a compound impractical rather than agent-like (halosilanes, sulfonyl halides, peroxy acids). Tier C records a simple phosphoryl ester without a leaving group and is noted rather than excluded. An earlier version that did not distinguish leaving groups flagged DMMP at the same tier as sarin and admitted potassium cyanide to the faithful-and-safe quadrant; both were corrected.

2.8. Reproducibility

All analyses are deterministic given the seed and the permutation test is exhaustive rather than sampled. Numerical work used NumPy 2.5.3 [59], SciPy 1.18.1 [60], pandas 3.0.5 [61] and scikit-learn 1.9.1, figures Matplotlib 3.11.2 [62], and descriptors RDKit 2026.03.6 [39]. Code, descriptor tables, parsed toxicity values with their sources and the figure scripts are in the repository cited under Data and code availability.

3. Results

3.1. What the designating documents say, and for which measurement

Searching for an instrument that designates the three incumbent compounds as chemical warfare agent simulants, we found that two carry a documented designation for a purpose other than the one they now serve, and the third carries none. The record separates into three grades, set out in Table 1: a compound named for a stated measurement, a compound recorded only as having been used, and a compound with no document at all.

DMMP was identified as a test simulant for canister and cartridge filtration against GB and HD in a NIOSH respirator programme [63], and appears again in the same document's candidate list for the GB and HD permeation study. That document also states the simulant information it provides is not to be used as a recommendation of any simulant, and the property at issue in both roles is breakthrough through a barrier rather than detection chemistry. DPGME appears as "a preferred G agent simulant" in a patent assigned to the United States Department of the Army [36], in a training composition formulated with a fluorescent dye and justified on ion-mobility detectability, so the application is field training rather than sensor validation. For dichloromethane we found no designating document of any kind: it is absent from the NIOSH document and from

the Man-in-Simulant Test programme, and present in this role, as far as we can establish, only in the research literature, where it is run as a chemical agent simulant alongside DMMP and DPGME on metal-oxide gas sensors [35]. That study puts a fourth compound in the same role, acetonitrile, which the panel used here classes as an interferent, so the literature usage does not resolve to a fixed set either.

Table 1: The documentary record for the compounds this paper treats as simulants. Standing is graded by what the document does: *designated* names the compound for a stated measurement, *recorded* notes that it has been used without naming a measurement to validate it against, and *none* means no instrument was found. The measurement each compound is used for now is detection by a sensor, except in the last block, which lists two compounds the same NIOSH document names for filtration and which neither panel carries; they are shown because the rule by which the primary panel was built would have admitted them.

Compound	Standing	Instrument	Measurement it was named for
Treated as incumbent simulants here			
DMMP	designated	NIOSH respirator programme [63]; also on that document's permeation candidate list	canister and cartridge breakthrough against GB and HD
DPGME	designated	US Army patent [36]	ion-mobility response in a training composition
dichloromethane	none	none found	none
Added to the primary panel			
methyl salicylate	designated	Man-in-Simulant Test [2, 3]	vapour penetration of a protective ensemble
DFP	recorded	NIOSH notes [63]	none stated
CEES	recorded	NIOSH reference tables [63]	barrier breakthrough against HD
DCP, DCNP	none	none found	none
Named by the same document, not in either panel			
carbon tetrachloride, cyclohexane	designated	NIOSH respirator programme [63]	filtration against GB and HD

A negative claim about documents is only as good as its scope. We searched the standard-setting and test-procedure literature reachable without a subscription or a clearance: NIOSH and NPPTL respirator documents, US Army and DEVCOM reports indexed by the Defense Technical Information Center, United States patents assigned to the Department of the Army, National Academies assessments of Department of Defense test programmes, ASTM and NATO standards catalogues, and the peer-reviewed simulant literature cited here, together with the US Army Test Operations Procedures reachable through the same index. That search has a hole in it, and the hole is the most important document of all. TOP 08-2-196, *Simulant Selection for Laboratory, Chamber, and Field Testing* [6], is the Army's procedure for exactly the question this section asks, and we could not obtain it. We know its title and its role only because a public procedure cites it: TOP 08-2-188 CN1, which governs chemical point detector vapour testing, defers simulant selection to it [5]. The same public procedure states a criterion that is about detection rather than about barriers, that the simulant "should produce a signature signal similar to the one from the threat analyte", so the field does have at least one written statement of what a detection simulant

is for, and we did not find it where we first looked. TOP 10-2-022A is public as well and names methyl salicylate for the suit-level vapour challenge [4]. National CBRN defence manuals are generally not public.

What follows from this is a limit on the claim rather than a retraction of it. We can say that the three compounds the sensor literature runs on carry no public designation naming a detection measurement, and that the designations we did find name other measurements. We cannot say that no such instrument exists, and TOP 08-2-196 is where one would be. A reader with access to it should treat this section as a question put to that document rather than as an answer about it. The measurements in the rest of the paper do not depend on how that question comes out.

Two observations follow. A designation is specific to a measurement: a compound chosen because it breaks through a carbon bed like sarin has been validated for adsorption and not for an optical or electrochemical response, and one chosen because it lights up an ion-mobility detector in a training lane has been validated for that detector. The sensor literature using these three is using designations issued for other purposes. The compounds with the clearest designations, in turn, are not these three, and the clearest of them is not clear in the way we first read it. Methyl salicylate is the simulant of the United States Army's Man-in-Simulant Test programme, and it is carried into a published consensus standard that fixes the measurement as vapour penetration of a protective ensemble [3]. The National Academies assessment of that programme, however, states that its selection "was based on its historical use as an agent simulant" rather than on a validation against the agents, asks "whether MeS is a reliable simulant of chemical agents", notes that the physicochemical properties of VX, sulfur mustard and methyl salicylate are very different, and records that the assumptions behind its use "have not been tested" [2]. So the compound with the strongest instrument behind it carries, in the same literature, an explicit statement that the instrument rests on prior use. That is the pattern of this section rather than an exception to it, and it is why Table 1 grades what a document does and not how authoritative it looks. The NIOSH document's candidate list for the permeation study names eleven compounds, among them DIMP, DEEP, TEP, di-n-butyl sulfide and DMMP itself. Diisopropyl fluorophosphate and methyl salicylate are not on that list; they appear in the same document's notes, each recorded as having been used as a vapour simulant rather than designated for a measurement. Malathion and 2-chloroethyl ethyl sulfide appear in its reference tables of barrier breakthrough data. The compounds named in that document and in the Man-in-Simulant Test are in the thirty-five-compound panel of Section 2.1, not the twenty-eight-compound one. A recent survey states the first half of this in general form, that simulants have generally been selected on the basis of their historical use [64]. That survey records the absence of a development standard for *explosive* simulants, not for chemical agent simulants, and we have found no source that states the latter; the absence we report here rests on our own search and not on a prior claim.

3.2. What the comparison contains decides the answer

On the thirty-five-compound panel, no authentic agent has a valid match to an incumbent simulant. Figure 1a shows why, and the mechanism is not the one our earlier version reported. The two panels do not share a scale, and they do not quite share a space. Standardisation and the normalising separation are recomputed on each panel, and the filters admit different columns on each, `kappa1` on the twenty-eight-compound panel against three phosphorus counts on the primary one, so the structural coordinates are neither the same set nor the same number, thirty-seven against thirty-nine. Distances across the two are therefore not directly comparable, and the comparison we rely on is made within the primary panel, where the compounds, the columns and the standardisation are fixed and only the labels move. Held on that single basis, every agent

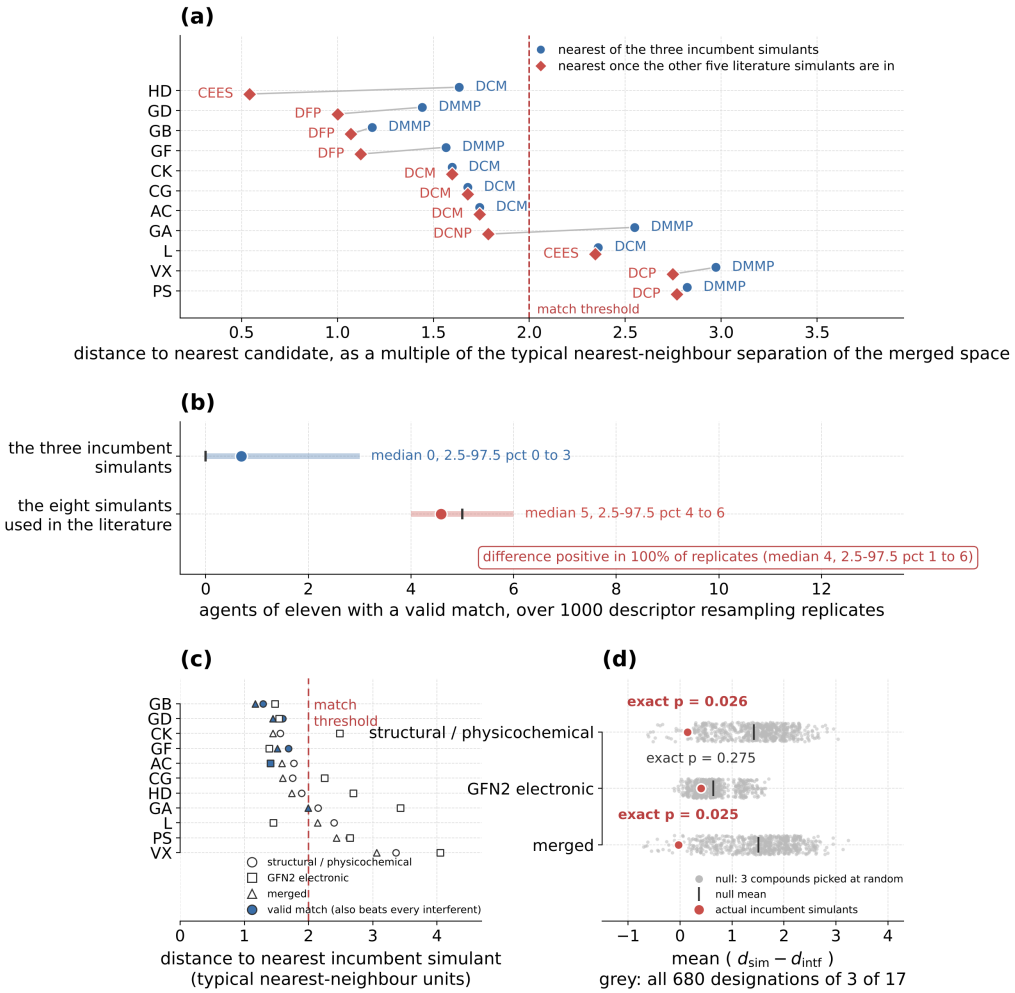


Figure 1: What the comparison contains decides the answer. (a) For each agent, the distance to the nearest incumbent simulant (circles) and to the nearest candidate once the five further compounds treated as simulants in the literature are included (diamonds), as multiples of the typical nearest-neighbour separation of the merged space on the thirty-five-compound panel. A match requires a normalised distance below the dashed threshold *and* a distance smaller than that to every compound not designated a simulant, so sarin, soman and cyclosarin fail on the primary panel despite lying inside the threshold. (b) Matched agents under 1000 descriptor-resampling replicates (Section 2.6) for the incumbent three and for the eight literature-used simulants examined here; circles are means, vertical rules medians, bars the 2.5th to 97.5th percentiles across replicates. (c) On the twenty-eight-compound panel, which omits those five, the distance from each agent to its nearest incumbent simulant in each of the three spaces, normalised by the typical nearest-neighbour separation of that space; filled markers are valid matches. (d) Exact permutation null for that panel over all 680 ways of designating three of the seventeen non-agents as simulants, with the actual designation marked; the p shown is the mean-difference statistic only, the other five and the adjusted values being in Supplementary Table S1, where the Benjamini-Hochberg value over its eighteen tests is 0.165.

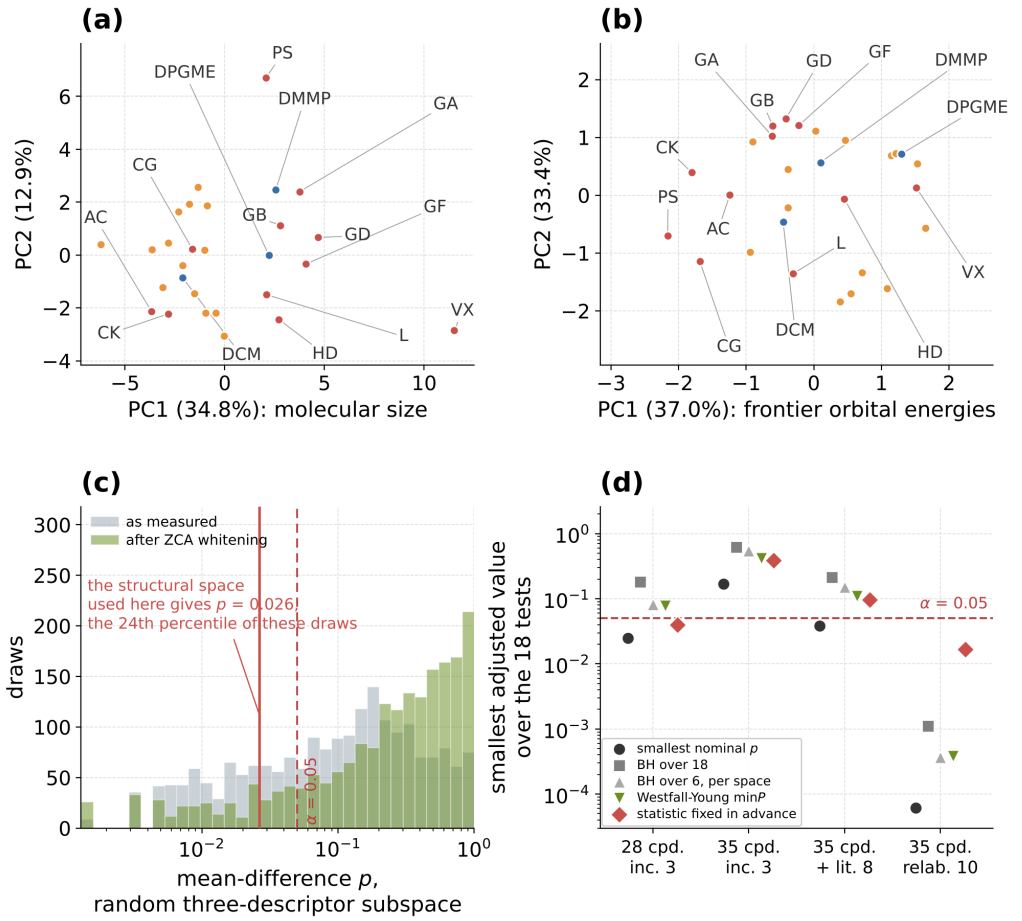


Figure 2: The spaces the comparison is made in, and what the nominal value is worth once the choices behind it are counted. Red marks the authentic agents, blue the three incumbent simulants and orange the interferents and solvents; only agents and simulants are labelled. (a) Principal components of the structural space and (b) of the three-coordinate GFN2-xTB electronic space, both on the twenty-eight-compound panel, so that the five literature simulants do not appear in them; the leading structural component is a size axis and the leading electronic component loads the two orbital energies against the dipole moment. (c) The mean-difference statistic for 2000 random three-descriptor subspaces of the structural space, before and after ZCA whitening, with the space used here marked: its 0.026 is the 24th percentile of those draws, and 36% of them fall below 0.05 as measured, 17% after whitening. (d) Smallest adjusted value over the eighteen tests of each configuration under the five accounting schemes of Supplementary Table S2; on the twenty-eight-compound panel only the scheme that fixes the statistic in advance falls below 0.05, and the fourth configuration falls below it under every scheme. Local reactivity at phosphorus and the rank of the Parr index are in Supplementary Figure S2.

sits exactly as far from its nearest incumbent whether the designated set is the incumbent three or those three together with the five literature compounds, because all three incumbents belong to both sets. The compounds did not become less faithful: on that panel sarin stands 1.18 from DMMP, soman 1.44 and cyclosarin 1.56. What changed is that diisopropyl fluorophosphate is present, at 1.07, 1.00 and 1.12 from the same three agents, and the match criterion requires a simulant to be nearer than every compound not designated one. The same holds across the panel, with 2-chloroethyl ethyl sulfide at 0.54 from sulfur mustard where dichloromethane stands at 1.67, and diethyl cyanophosphonate at 1.79 from tabun where DMMP stands at 2.55. Admitting the five as simulants gives five matched agents of eleven, against four on the twenty-eight-compound panel and zero on the primary one. The zero does not depend on the threshold: no incumbent simulant is the nearest non-agent compound to any of the eleven agents, so the count stays at zero however far the threshold is relaxed.

That count is sensitive to the descriptors, and the honest statement of it is the interval rather than the point. Resampling the 42 merged-space descriptors with replacement (Section 2.6) gives the incumbent three a median of zero matched agents with a 2.5th to 97.5th percentile range of zero to three, and exactly zero in 58.6% of replicates, against a median of five and an interval of four to six for that eight-compound set, which contains the incumbent three. Because the match criterion asks whether a designated compound is nearer than every compound outside the designated set, enlarging that set cannot lower the count, and part of the difference is set size alone: designating eight non-agents at random matches 2.68 agents on average against 1.00 for three, so the observed gap of five decomposes into 1.68 expected from size and 3.32 beyond it. Matching on size alone is not quite the right null, because the observed eight-compound set contains the incumbent three while a set drawn freely need not. Holding the three fixed and drawing the other five at random gives a mean of 1.92 rather than 2.68, and the observed five sits at $p = 0.068$ against it. Neither observed count departs from a null built to resemble it: the three at $p = 0.479$ over all 2024 designations, and the eight at $p = 0.068$ conditionally or $p = 0.161$ unconditionally. We report the lower tail for the three because the observed count is zero, which is the floor; the upper tail there is 1.000 by arithmetic and says nothing, and an earlier version of this paper quoted it as though it did. The lower tail carries its own limit, and it belongs here rather than in a footnote: because zero is the smallest count attainable, $p = 0.479$ is the smallest lower-tail value this comparison can ever return. It does not report a result that failed to reach significance. It reports a comparison that cannot reach significance at any conventional level, whatever the data. What the resampling adds is that the difference between the counts, unadjusted for size, is positive in 1000 of 1000 replicates, median four, interval one to five (Figure 1b). The per-agent reliabilities in the same replicates separate the firm results from the marginal ones. Sulfur mustard is matched by 2-chloroethyl ethyl sulfide in every replicate and tabun by diethyl cyanophosphonate in 51.2%, while under the incumbent set alone sarin recovers DMMP as its nearest compound in 38.0% of replicates, soman in 14.9% and cyclosarin in 11.4%. The displacement of DMMP at the G agents is therefore the weakest part of the picture and the sulfur mustard result the strongest, reversing what the point estimates suggest.

Figure 2a and b give the spaces on the twenty-eight-compound panel. The leading structural component is a size axis carrying 34.9% of the variance and the leading electronic component loads on the two orbital energies against the dipole moment, the first two carrying 70.4%. The primary panel behaves the same way, at 34.9% and 70.3%. Merging the two blocks returns a size axis at lower cumulative variance over the first two components, 45.7% on the smaller panel and 45.2% on the primary one, for a dimensional rather than a chemical reason: at equal per-descriptor weighting the structural descriptors, thirty-seven on the smaller panel and thirty-nine on the

primary one, outvote three electronic ones.

Every exact permutation test we can run, across four panel and designation configurations, three spaces and six statistics, is in Supplementary Table S1, in full rather than selected from, because the pattern across the cells is the finding. Three things hold across it. On the primary panel no test reaches nominal significance: in no space and under no statistic are the incumbent simulants distinguishable from compounds drawn at random. Classing the five literature compounds as simulants recovers two nominal values below 0.05, both in spaces containing the structural descriptors, so the difficulty is not that the test cannot detect anything on this panel. On the twenty-eight-compound panel, which differs only in omitting those five and two organophosphorus pesticides, the incumbents beat random compounds under two of six statistics in the structural space and three of six in the merged space.

Correcting for the number of times we looked removes almost all of it, and the exception is instructive. On the primary panel the question does not arise, since the smallest nominal value is already 0.156 under the per-descriptor standardisation used throughout. The five accounting schemes of Supplementary Section S6, applied in Supplementary Table S2, matter only for the twenty-eight-compound panel. There the eighteen-test family gives a smallest adjusted value of 0.181, correcting the six statistics within each space gives 0.074, and the Westfall-Young min-P procedure, which uses the shared permutations and so accounts for the dependence between the tests exactly rather than approximating it, gives 0.078. That is the closest any correction brings the panel to the conventional threshold without crossing it. The fifth scheme behaves differently. Had the mean difference been specified in advance as the single statistic, three tests would have been run and the twenty-eight-compound structural and merged results would survive at $q = 0.040$, while neither thirty-five-compound configuration would (0.386 and 0.092). We did not specify a statistic in advance; we computed six and chose among them after seeing them, which is the circumstance the correction is for. The comparison is the paper's own argument made with its own data: one analysis, one data set, five defensible ways of accounting for the search, and a result significant under one and not the other four. The declaration that would have made the difference is one nobody in this literature makes, and it has to be made before the data are seen.

One label decision carries more than any of this. Section 2.1 classes dichlorvos and malathion as interferents and records that decision as contestable. Classing them as simulants instead, which moves two labels and changes nothing else, gives a smallest nominal value of 0.00006 and a Benjamini-Hochberg value of 0.0011 over the same eighteen tests, and it is the only configuration in this work that survives every one of the five accounting schemes of Supplementary Table S2. The direction is not surprising, since both pesticides carry the phosphoryl motif the agents carry, and that is the point: significance here is available to an analyst who moves two labels, and the practice records neither which labels were moved nor that any were. This is not a recommendation, and the reason it is not is narrower than we first wrote. We described the two pesticides as compounds nobody uses as simulants. That is wrong for one of them: the NIOSH document cited throughout this section lists malathion in its GB simulant column and reports breakthrough times for it through protective clothing materials [63]. Dichlorvos we still find in no such role. So the relabelling that produces the only result surviving every accounting scheme is not an arbitrary one, which strengthens rather than weakens the point: a defensible label change, made by an analyst with a citation in hand, moves this analysis from nothing at $q = 0.617$ to $q = 0.0012$, and the practice records neither that a label was moved nor which one. What the comparison shows is that a designated set can be made to look supported by choosing what goes into it.

One further observation bounds what the nominal values can mean. The count of agents for

which a compound outside the designated set is nearer than any incumbent simulant is 11 of 11 in the merged space of the primary panel against a null expectation of 9.63 set by the 21:3 group-size asymmetry alone, and 6, 10 and 6 of 11 in the three spaces of the smaller panel against 9.06, so the raw count carries no evidence without the permutation test and in one space runs the wrong way.

The difference between the two spaces is chemical rather than geometric. Truncating the standardised structural space to its first three principal components strengthens the result rather than weakening it, from $p = 0.026$ to $p = 0.0029$ on the twenty-eight-compound panel, which leaves the two spaces 94-fold apart in p at matched dimension. What the nominal structural value is worth against the descriptors it was drawn from is a separate question, and there the answer has two parts that point in opposite directions, and an earlier version of this paper reported only the first. Of 2000 random three-descriptor subsets of that space, the observed 36-dimensional value of 0.026 sits at the 24.3rd percentile of the resulting distribution while the electronic value of 0.275 sits at the 75.5th: the full-space value is an ordinary draw among the descriptor subsets we could have used instead, which is a statement about descriptor choice and is the one we rely on.

The second part is that 36.0% of those subsets reach $p < 0.05$, and we first read that as showing the same thing. It does not. In a fixed subspace the permutation p is the rank of the designated set among all 680 designations, so if the designation were arbitrary the expected share below 0.05 would be 0.05. We therefore recomputed the share for every one of the 680 designations over the same subspaces (Section S11). The mean is 4.7%, as it should be; the incumbent designation's 36.0% ranks fifth of 680, $p = 0.0074$. The structural block therefore does carry something that distinguishes the incumbents from an arbitrary triple, and we withdraw the claim that this control shows otherwise.

What it carries is narrower than it sounds, and the sign of the statistic says so. The observed mean difference in the structural space is +0.155: the designated simulants are on average *further* from the agents than the nearest compound outside the designated set. What the test detects is that they are less far than a triple drawn at random from a pool assembled out of interferents and solvents, which were chosen for this panel because they are not like the agents. Against that reference class a single organophosphorus compound is close to sufficient. Supplementary Section S10 reports both controls in full.

3.3. How many agents have a valid simulant depends on the panel

Table 2 lists, for each agent on the twenty-eight-compound panel, the nearest incumbent simulant in the merged space as a multiple of that space's typical nearest-neighbour separation. Sarin (1.171), soman (1.445), cyclosarin (1.449) and tabun (1.905) are matched, all four by DMMP; VX at 2.922 is not. Tabun clears the threshold of 2 by 0.095, and we flag that rather than report it as a result: an earlier version of this analysis, which retained a structural coordinate that is an exact function of two others (Section 2.2), put tabun at 2.003 and counted three matches here instead of four. A count that moves when a redundant coordinate is removed is a count that should be read as marginal, which is the same lesson the rest of this paper draws about the panel. DPGME is nominally VX's nearest simulant in all three spaces, but at 3.060 typical separations in the merged space and 3.357 in the structural space that is not a match in any useful sense. Dichloromethane, which the sensor literature we surveyed runs against the whole panel rather than against a named agent [35], is not phosgene's nearest neighbour in any of the three spaces: ethyl acetate leads it in the structural space (1.136 typical separations against 1.750) and in the merged space (1.120 against 1.600), and chloroform in the electronic space (1.171 against 2.247).

The direction of the comparison matters. Table 2 gives, for each agent, its nearest simulant; the reverse question is which agent each simulant is nearest to, and over the same 1000 descriptor-resampling replicates of Section 2.6 the three incumbents separate. DMMP is nearest sarin in 92.8% of replicates and nearest one of the four G-series agents in 99.5%. Dichloromethane is nearest a G-series agent in 4.2%, sitting instead with the choking and blood agents. DPGME divides evenly, sulfur mustard in 38.2% and the G series in 56.9%. Being nearest is not being close, and DPGME carries no valid match in any space on this panel; but it is the one of the three whose designating document names a class, "a preferred G agent simulant" [36], and it is the one whose class these spaces do not resolve.

Table 2: Nearest incumbent simulant for each agent on the twenty-eight-compound panel, as a multiple of the typical nearest-neighbour distance of the merged space. A match requires a normalised distance ≤ 2 together with a distance smaller than that to every compound outside the designated set. On the primary thirty-five-compound panel no agent has a match.

Authentic agent	Nearest simulant	Normalised distance	Valid match
GB	DMMP	1.171	yes
GD	DMMP	1.445	yes
CK	DCM	1.446	no
GF	DMMP	1.517	yes
AC	DCM	1.587	no
CG	DCM	1.600	no
HD	DCM	1.740	no
GA	DMMP	1.995	yes
L	DCM	2.141	no
PS	DMMP	2.433	no
VX	DPGME	3.060	no

The count is stable but not invariant, and the two spaces admit tabun at different bars. In the merged space the set is {GB, GD, GF} at thresholds of 1.5 and 1.75 and {GB, GD, GA, GF} at 2.0, 2.5 and 3.0, tabun entering once the bar passes its value of 1.995. In the structural space it is {GB} alone at 1.5, with soman at 1.600 and cyclosarin at 1.687 outside, {GB, GD, GF} at 1.75 and 2.0, and four only at 2.5. The four-agent figure is therefore a statement about the merged space at a threshold of exactly 2.0, and the three-agent figure the same space one notch lower. The electronic space is the exception, and it is the informative one: there the count is one at every threshold tested, hydrogen cyanide matched by dichloromethane at 1.407, because acetonitrile sits closer to sarin, soman and cyclosarin than DMMP does and all three fail the threshold-free condition outright. Over the thresholds tested the count does not exceed four of eleven in any space.

The dependence on the panel is sharper than the dependence on the threshold. On the thirty-five-compound panel, with the incumbent three still the only simulants and the five literature compounds in the comparison pool, the merged space returns no valid match for any of the eleven agents: DMMP loses GB, GD and GF to diisopropyl fluorophosphate, which is nearer to all three. Admitting the five as simulants gives five matches of eleven, more than the incumbent set achieves on either panel, with DFP taking sarin, soman and cyclosarin at 1.06, 0.99 and 1.11, diethyl cyanophosphonate taking tabun at 1.79, and 2-chloroethyl ethyl sulfide taking sulfur mustard at 0.57, the closest agent-simulant pair anywhere in this work. VX remains unmatched in every

panel and space we examined.

We draw no conclusion from this that the field should switch compounds, because the five were not selected by any procedure we can inspect either, and Section 3.2 applies to them exactly as it applies to the incumbent three. What the comparison establishes is narrower: the statement "three of eleven agents have a valid simulant" is true of one panel, false of a second, and replaced by a different list on a third, which is the same instability the permutation tests report and not an independent finding.

One agent deserves a note of its own. Lewisite's incumbent simulant is dichloromethane, which shares neither its central element nor its halogen count, so the failure might be read as an artefact of that comparator rather than of the designation. Arsenic trichloride, used as a lewisite simulant in the sensing literature [65], though we found no designating document for it, leaves the incumbent-three count at zero of eleven and supplies lewisite's only match under the literature set, at 1.25 where the best the five could previously offer was 2-chloroethyl ethyl sulfide at 2.35 (Supplementary Information).

3.4. What the array actually measures

Every argument above rests on descriptor distances, and the premise that one of those spaces corresponds to what a vapour-phase sensor reports is tested in Supplementary Section S16 against measured colorimetric response for twelve panel compounds over 29 dyes, of which 23 are independent channels in the one plate set where that matters. It does not resolve the question, and it is worth being exact about why. Of twelve panel-channel-space cells, three reach a nominal $p < 0.05$ under Mantel permutation over compounds and they do not agree on a space; corrected across the twelve, the smallest adjusted value is 0.100 and no cell reaches $q < 0.05$. All three of those cells lie in the plate set that contains no simulant, so they compare one agent with another and not a simulant with an agent. The six cells that bear on the premise are the six in the set that holds DMMP, and all six are null. Simulating the test actually run, the exact Mantel over all 720 relabellings of six compounds, the design detects $|\rho| \geq 0.67$ at 80% power, so these data exclude a strong association and nothing weaker. The premise is therefore untested rather than supported, and every distance-based claim in this paper inherits that. Three artifacts are reported there because our own first version contained all three: dye-axis centring fixes the between-compound mean cosine at $-1/(n-1)$, which the observed means clear by only 0.013 to 0.064; six dyes that do not fluoresce under ultraviolet illumination returned one background reading six times, which we now collapse; and the measured agent nearest to DMMP in the ultraviolet channel is tabun rather than sarin, which the descriptor criterion of Section 3.3 excludes.

3.5. The toxicity half of the criterion

The criterion has a second half: that compounds chosen on physicochemical grounds present minimal toxicity risk [21]. We tested it on the same panel and report it in full in the Supplementary Material. Three results are needed here.

Dichloromethane has a regulator-derived acute inhalation value and DMMP has neither an acute one nor a chronic one: it is absent from the AEGL list, and EPA's provisional assessment separately declined to derive a chronic inhalation reference concentration for want of data (Figure 3a and b, Supplementary Section S1). The two incumbents therefore differ in how well documented they are and not only in how hazardous, which matters because DMMP is also the compound against which decontamination efficiency is quantified [1]. Dichloromethane's value is favourable on the acute axis, at the 96th percentile of a 207-compound pool, and says nothing about the chronic and carcinogenic classification now displacing it from solvent inventories.

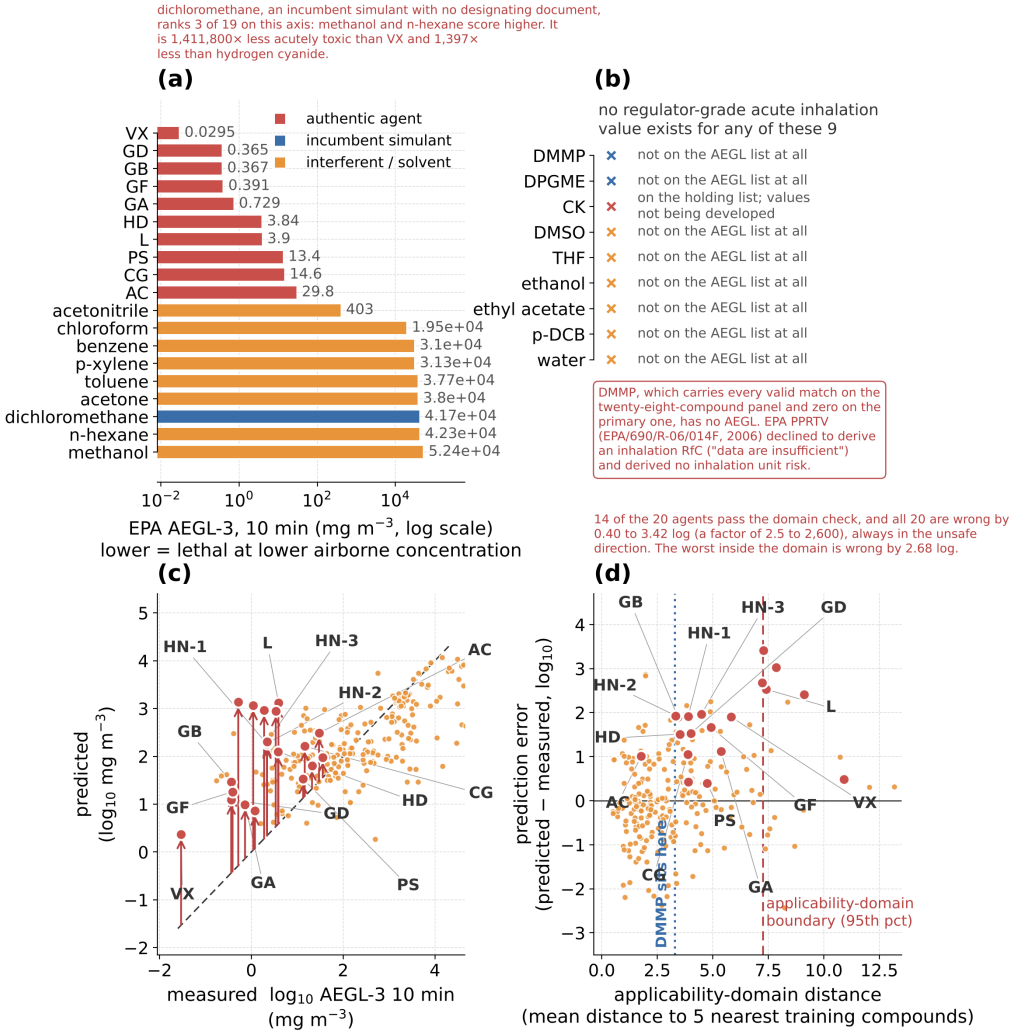


Figure 3: The second half of the criterion, measured and then modelled. (a) EPA AEGL-3 at 10 min on a logarithmic scale for the nineteen compounds that have one; the incumbent simulant dichloromethane sits near the safe end but is not the safest, methanol and n-hexane both scoring higher. Five of the nineteen values come from the agency's own footnotes rather than from the table cell, because they exceed half the lower explosive limit. (b) The nine compounds with no regulator-grade acute inhalation value, with the reason for each; DMMP, which supplies every valid match on the twenty-eight-compound panel and zero on the primary one, is among them. (c) Out-of-fold predictions for the 230 training chemicals (orange) with the twenty military agents held out of training (red) overlaid; arrows run from the measured value to the prediction, and every agent is predicted safer than it is. Labels name the ten agents of this paper's panel that the AEGL set also holds and the three nitrogen mustards; the remaining seven held-out agents are plotted unlabelled. (d) Prediction error against applicability-domain distance. Every one of the twenty is wrong by between 0.40 and 3.42 log units, and fourteen of them pass the domain check, the worst of those fourteen being wrong by 2.68 log units. The dotted line marks where DMMP sits.

A model cannot fill the gap. A random forest trained on 230 AEGL chemicals reaches $R^2 = 0.458$ and then predicts every one of twenty held-out military agents safer than measured, by factors of 2.5 to 2600, while certifying fourteen of them as inside its applicability domain (Figure 3c and d, Supplementary Section S2). The failure is not a missing mechanism, since thirteen training compounds carry an acetylcholinesterase-inhibiting pharmacophore and one is more acutely toxic than sarin. It is tail regression, a bias in the response space that an input-space domain check cannot see.

Fidelity and acute toxicity show no monotonic association across the 207 compounds carrying both measured axes (Spearman $r = +0.062$, 95% CI -0.077 to $+0.199$). The interval excludes a trade-off strong enough to constrain selection, which is weaker than independence and is what the data support, so the two can be improved separately (Supplementary Section S19). We draw no candidate recommendation from that: one resting on a single panel and two axes would repeat the error this paper reports.

4. Discussion

The criterion by which low-toxicity simulants are chosen is stated clearly enough to test [21]: align hydrophobicity, volatility and solubility with the agent. What the practice does not state is what the chosen compounds should be compared against, and that decides the answer. On a panel containing the compounds the field itself uses as simulants, the three incumbents match no agent and no test reaches nominal significance; on a panel omitting them, four agents are matched and the smallest nominal value is 0.025.

The mechanism is not a story about fidelity. The incumbents sit almost as close to the agents on one panel as on the other, and on a single standardisation basis they do not move at all (Section 3.2). What the larger panel supplies is diisopropyl fluorophosphate, nearer to sarin, soman and cyclosarin than DMMP is, so a criterion that asks a simulant to beat the alternatives returns nothing. Resampling the descriptors leaves the aggregate comparison intact in every replicate, but not each displacement within it: sulfur mustard is matched by 2-chloroethyl ethyl sulfide in all of them while the incumbent set recovers DMMP at sarin in 41.0%, soman in 14.4% and cyclosarin in 11.4%, so the G-series displacement is the weakest part of this and the mustard result the strongest. A designation defended by a comparison that excludes the alternatives is not defended.

An earlier version of this work reported the twenty-eight-compound structural cell as the finding and built a thesis on the contrast between it and the electronic space. Three things undo that reading, and each is a mistake another group could make the same way. The value does not survive the eighteen tests of its own family (Benjamini-Hochberg $q = 0.181$), although it would have survived at $q = 0.040$ had the statistic been fixed in advance, which demonstrates the paper's argument rather than defending the value. It sits at the 24.3rd percentile of 2000 arbitrary three-descriptor slices of the same block, so the particular space we used is not what produced it. And it is specific to the panel that omits the five. What does not undo the reading, and what we reported as though it did, is the share of those slices reaching $p < 0.05$: corrected against the 680 designations that share generates, the incumbent designation ranks fifth (Section 3.2).

Two things we expected to dissolve the contrast between the spaces did not. The Parr electrophilicity index was an exact function of two of the other electronic descriptors, so the electronic space never had the dimension we reported; removing it moved that space's value only from 0.297 to 0.275. Matching the dimensions did not dissolve the contrast either (Section 3.2). What does weaken it is the choice of electronic-structure method. Rebuilding all three coordinates

of the electronic space at the density-functional level moves it from 0.275 to 0.0368 and from one valid match to five, four of them pairings the designation record supports; the merged space, rebuilt the same way, holds at 0.0221 (Section S5). We therefore withdraw the contrast as something to build on: it is not a stable feature of the comparison, and the structural value anchoring it is in any case an ordinary draw from the subspaces we could have chosen.

What remains is a claim about comparison rather than about chemistry. On the primary panel we cannot establish that the incumbent simulants are closer to the agents than compounds drawn at random, and we cannot establish the opposite either. What we can establish is that the compounds which would make the comparison informative were not in it.

The documentary record explains how that arose, and it is not a story about carelessness. Two of the three were designated carefully for measurements that are not the one they are now asked to support (Section 3.1): DMMP for adsorptive breakthrough on a filter bed [63], DPGME for ion-mobility detectability in a training composition [36]. Each selection was fit for its own measurement. The compounds then carried their status into sensor development, where the governing property is the response of a dye or an electrode rather than a breakthrough time, and nothing in the original validation speaks to that. For dichloromethane we found no designating document at all.

The absence this paper documents is specific rather than general. A safety-screened repository of candidate simulants exists [26], and a descriptor-based selection procedure appeared in this journal fifteen years ago [23]. What is missing is any route by which either reached the compounds the detection literature settled on.

That mechanism is general. A simulant is validated against a measurement rather than against an agent, and treating the validation as a property of the compound, so that it travels into a new application, produces a working set whose standing no descriptor space can reproduce. The corrective is not a replacement compound but a declaration: which measurement a simulant was validated against, and a panel, space and statistic fixed before a claim that it is valid for a different one is tested. Agrawal et al. [22] reach the same place from the other direction, adequacy being a property of the compound together with the measurement it stands in for.

Three consequences follow for practice. DMMP is a surrogate for one G agent at a time rather than for the G series, and not a V-agent surrogate at all: the descriptor criterion assigns GB, GD and GF to it, while the measured array puts sarin first in reflectance and tabun, which fails that criterion, first in the ultraviolet channel (Supplementary Section S16). The condensed Fukui index at phosphorus puts DMMP above VX by a factor of 2.7 in the semi-empirical treatment used here, which would sharpen the point, but we could not reproduce that index at a higher level of theory and we do not lean on it (Supplementary Section S5). The global electrophilicity index should not enter a descriptor space, for an arithmetic rather than a chemical reason: where the frontier orbital energies are present it adds no information, and a correlation filter will not notice. It should not be read as a reactivity axis either, and that is a chemical reason: recomputing this panel with density functional theory leaves the qualitative picture but moves which agents sit lowest, so an ordering taken from it describes the method (Supplementary Section S15). And the safety of a simulant is a question about evidence before it is one about hazard, since dichloromethane carries a regulator-derived acute inhalation value and DMMP carries none (Section 3.5).

We did not attempt to improve the toxicity model by training on organophosphorus compounds, and the distinction is narrower than a refusal to model the chemistry. Mechanistic structure–activity models of organophosphorus lethality exist and we have contributed to them [34]; such a model assesses compounds that already exist. What we decline to build is the same instrument turned into a design objective, a predictor of mammalian lethality evaluated over structures not yet made

inside a loop that proposes them, which is the configuration in which a safety model becomes a potency model [32]. The cost is an acknowledged gap: for the compound class of greatest interest we can say that a predicted value is unreliable and cannot supply a better one. Reporting the gap rather than closing it is the correct outcome, and where to set that boundary is contested rather than settled [33].

Limitations.. The documentary findings of Section 3.1 rest on public sources, and the scope of that search is stated there. The claim is that no public instrument designates these three compounds for the use the sensor literature makes of them, not that no instrument does. A null result invites the question whether the design could have seen an effect. On the primary panel the mean-difference statistic reaches $p < 0.05$ once the incumbent simulants are moved 0.22 typical nearest-neighbour separations closer to the agents, while the deficit they actually carry has a median of 0.445 and falls below that value only at sarin, where it is 0.110 (Supplementary Section S14). The test is therefore sensitive to effects smaller than the one at issue. The agent set is eleven compounds and the nearest-neighbour statistics inherit that: on the larger panel with eight simulants, leaving out one agent at a time moves the structural p between 0.020 and 0.186, the worst case being removal of tabun, so a result one of eleven agents can carry across the 0.05 line is not a property of the panel. The permutation test is directional and reported one-sided, with both two-sided conventions given in Section 2.4. The two panels are the two we could build from compounds carrying complete descriptor data; measured toxicity values exist for a subset only, DMMP among the compounds without one, so the two axes do not run over the same compounds. A fourth configuration, classing dichlorvos and malathion as simulants, is reported in Supplementary Table S1 and Section 3.2 because we flagged that label decision as contestable; the four configurations sample the configuration space rather than census it. Accordingly, the present panels should be interpreted as demonstrations of panel dependence rather than as exhaustive representations of all compounds that have been used as chemical warfare agent simulants. One omission is worth naming because it is systematic. Probe chemistries for lewisite are developed against arsenic trichloride [65], which is in neither panel, and the structural filter of Section 2.7 flags every arsenical, so a compound of that class could not have been admitted as a candidate in the toxicity screen either. The arsenical route to a lewisite simulant is therefore outside what is tested here, and lewisite's lack of a valid match in any space should be read against that. The measured-array intervals describe variation across concentrations within one plate set rather than agreement between preparations (Supplementary Section S16). GFN2-xTB is semi-empirical, and the quantities used here are relative within a consistent protocol rather than absolute. The virtual orbital energies make that concrete. Across the thirty-five-compound panel they run from −9.64 eV for chloropicrin to +2.23 eV for water, and twenty-four of the thirty-five fall below −4 eV. Read as electron affinities in the Koopmans sense those values are not attainable, chlorine being the most electron-avid element at 3.6 eV, so the method's virtual eigenvalues are coordinates and not affinity estimates. Every distance reported here is computed on descriptors standardised within the panel, which leaves a shift and a scale applied to a whole axis without effect. What such an axis cannot carry is a claim about any one compound's absolute frontier energy, and we make none.

5. Conclusions

Simulant validity cannot be treated as an intrinsic property of a compound. It depends on the measurement it is intended to reproduce and on the comparison against which that claim is

evaluated, and the practice of choosing chemical warfare agent simulants records neither. On a panel of thirty-five compounds that includes the five the sensor, sorbent and decontamination literature also uses as simulants, not one of eleven authentic agents has a valid match to an incumbent simulant and not one of eighteen tests reaches nominal significance. The incumbents are no further from the agents on that panel than on the smaller one; diisopropyl fluorophosphate is simply nearer, and under descriptor resampling the incumbent three match a median of zero agents against five for the eight literature-used simulants examined here, the difference positive in every one of 1000 replicates. Omitting those five restores four matches and a nominal structural p of 0.026, a value at the 25.9th percentile of 2000 arbitrary three-descriptor subspaces of the same block which no correction over the eighteen tests run leaves significant.

The documentary record gives the reason. DMMP was designated for filtration breakthrough testing and DPGME for ion-mobility training compositions, neither of which is detection chemistry; for dichloromethane we found no designating instrument at all. A simulant is validated against a measurement, and that validation does not travel with the compound into a new application.

The absence of evidence is not evidence of unsuitability, and we draw no conclusion that the incumbents are poor choices. What we conclude is narrower and addressed to method. A simulant claim is testable, this field has the data to test it, and the test returns whatever the analyst's undeclared choices imply. Moving two labels, from interferent to simulant, is enough to turn a result that survives no correction into one with an adjusted value of 0.0011. The practical form of that conclusion is a reporting rule: a simulant result should name the measurement the compound was validated against and the comparison set it was tested in, and should fix the panel, the descriptor space, the statistic and the descriptor weighting before the test is run, and should state how any conformer-dependent descriptor was taken. We add the last of those because we had not counted it ourselves until a reproduction check found it (Section S4); unlike the other four it does not move the answer here, but nothing in the design guaranteed that. Until those are declared in advance, a statement that a compound is a validated simulant reports a convention rather than a measurement.

Findings from the toxicity axis are in the Supplementary Material and do not depend on the contested comparison. The two incumbents differ in how well documented they are and not only in how hazardous, dichloromethane carrying a regulator-derived acute inhalation value and DMMP carrying none of any kind (Section 3.5). A random forest trained on 230 AEGL chemicals predicts all twenty held-out military agents safer than measured, by factors of 2.5 to 2600, so predicted toxicity cannot stand in for the missing values (Supplementary Section S18). Fidelity and acute toxicity show no monotonic association across the 207 compounds carrying both axes (Supplementary Section S19). We propose no replacement compound, because proposing one would repeat the error this paper is about.

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CRediT authorship contribution statement

Ku Kang: conceptualisation, methodology, software, formal analysis, investigation, data curation, visualisation, writing – original draft. **Jeongyun Kim:** formal analysis, investigation,

validation. **Miae Kim:** investigation, validation, writing – review and editing. **Doo-Hee Lee:** validation, resources, writing – review and editing, supervision. **Jin Yoo:** methodology, investigation, validation, writing – original draft. **Yoon Jeong Jang:** validation, writing – review and editing, supervision, project administration.

Declaration of competing interest

The authors declare no competing financial interest.

Declaration of generative AI and AI-assisted technologies

During the preparation of this work the authors used a large language model (Anthropic Claude) to assist in drafting and editing prose, in writing and auditing the analysis and figure scripts, and in building the internal-consistency checks described under Reproducibility. No numerical result was produced by the model: every value reported here is read from the output of the deposited scripts, which were executed to generate it. The authors reviewed and edited all content and take full responsibility for it.

Data and code availability

Descriptor tables, parsed AEGL values with their sources, permutation distributions, candidate tables and all analysis and figure scripts are in a public repository at `https://github.com/bisu9082/simulant-designation-evidence`, at the tagged release `v1.0`, and are also provided with this article. The repository also carries the primary EPA source document from which every toxicity value here was parsed, so the parse can be checked against the original. The exact permutation test is exhaustive and the remaining analyses are deterministic given the stated seed. The descriptor cache is checked against the code rather than taken on trust: `build_descriptors.py -verify` rebuilds every numeric column and the analysis pipeline stops if any column fails to reproduce. Section S4 reports what that check caught.

References

[1] H. S. Chi, J.-m. Park, J. Yoo, G. J. Choi, S. H. Hong, J. Son, J.-G. Lee, K. Kang, Feasibility study of electron beam irradiation for decontamination of chemical and biological warfare agent simulants: Dosimetric and operational evaluation, Applied Radiation and Isotopes 232 (2026) 112582. doi:10.1016/j.apradiso.2026.112582.

[2] National Research Council, Technical Assessment of the Man-in-Simulant Test Program, The National Academies Press, Washington, DC, 1997. doi:10.17226/5936.

[3] ASTM International, Standard test method for man-in-simulant test (MIST) for protective ensembles, ASTM F2588-12(2020), designation F2588-12, reapproved 2020 (2020).

[4] US Army Test and Evaluation Command, Test operations procedure (top) 10-2-022a: Chemical vapor and aerosol system-level testing of chemical/biological protective suits, Tech. Rep. TOP 10-2-022A, US Army Test and Evaluation Command (Dec. 2013).

[5] US Army Test and Evaluation Command, Test operations procedure (top) 08-2-188 cn1: Chemical point detector vapor testing, Tech. Rep. TOP 08-2-188 CN1, US Army Test and Evaluation Command, approved for public release; Defense Technical Information Center AD1072837 (2019).

[6] US Army Test and Evaluation Command, Test operations procedure (top) 08-2-196: Simulant selection for laboratory, chamber, and field testing, Tech. Rep. TOP 08-2-196, US Army Test and Evaluation Command, cited in TOP 08-2-188 CN1 as the governing simulant-selection procedure; the document itself was not obtainable for this work (Apr. 2011).

[7] K. Kim, O. G. Tsay, D. A. Atwood, D. G. Churchill, Destruction and detection of chemical warfare agents, Chemical Reviews 111 (9) (2011) 5345–5403. doi:10.1021/cr100193y.

[8] Y. J. Jang, K. Kim, O. G. Tsay, D. A. Atwood, D. G. Churchill, Update 1 of: Destruction and detection of chemical warfare agents, Chemical Reviews 115 (24) (2015) PR1–PR76. doi:10.1021/acs.chemrev.5b00402.

[9] S. Royo, R. Martínez-Máñez, F. Sancenón, A. M. Costero, M. Parra, S. Gil, Chromogenic and fluorogenic reagents for chemical warfare nerve agents' detection, Chemical Communications (46) (2007) 4839. doi:10.1039/b707063b.

[10] M. R. Sambrook, S. Notman, Supramolecular chemistry and chemical warfare agents: from fundamentals of recognition to catalysis and sensing, Chemical Society Reviews 42 (24) (2013) 9251. doi:10.1039/c3cs60230c.

[11] L. M. Eubanks, T. J. Dickerson, K. D. Janda, Technological advancements for the detection of and protection against biological and chemical warfare agents, Chemical Society Reviews 36 (3) (2007) 458. doi:10.1039/b615227a.

[12] N. A. Rakow, K. S. Suslick, A colorimetric sensor array for odour visualization, Nature 406 (6797) (2000) 710–713. doi:10.1038/35021028.

[13] H. Lin, M. Jang, K. S. Suslick, Preoxidation for colorimetric sensor array detection of VOCs, Journal of the American Chemical Society 133 (42) (2011) 16786–16789. doi:10.1021/ja207718t.

[14] J. R. Askim, M. Mahmoudi, K. S. Suslick, Optical sensor arrays for chemical sensing: the optoelectronic nose, Chemical Society Reviews 42 (22) (2013) 8649. doi:10.1039/c3cs60179j.

[15] J. Kim, K. Kang, M. Shin, S. Kim, J. Yoo, M.-K. Kim, W. B. Lee, D.-H. Lee, Machine learning-enhanced colorimetric sensor array for rapid detection of nerve agents, Journal of Hazardous Materials 501 (2026) 140703. doi:10.1016/j.jhazmat.2025.140703.

[16] S. Bae, K. Kang, Y. K. Kim, Y. J. Jang, D.-H. Lee, Field-deployable real-time AI system for chemical warfare agent detection using YOLOv8 and colorimetric sensors, Chemometrics and Intelligent Laboratory Systems 261 (2025) 105365. doi:10.1016/j.chemolab.2025.105365.

[17] M. Shin, K. Kang, J. Kim, S. Kim, J. Yoo, Y. J. Jang, M.-K. Kim, E. V. Anslyn, D.-H. Lee, Portable machine learning–integrated optical sensor array for rapid and selective detection of authentic Novichok nerve agents, Sensors and Actuators B: Chemical 457 (2026) 139706. doi:10.1016/j.snb.2026.139706.

[18] S. L. Bartelt-Hunt, D. R. U. Knappe, M. A. Barlaz, A review of chemical warfare agent simulants for the study of environmental behavior, Critical Reviews in Environmental Science and Technology 38 (2) (2008) 112–136. doi:10.1080/10643380701643650.

[19] J. E. Mondloch, M. J. Katz, W. C. I. III, P. Ghosh, P. Liao, W. Bury, G. W. Wagner, M. G. Hall, J. B. DeCoste, G. W. Peterson, R. Q. Snurr, C. J. Cramer, J. T. Hupp, O. K. Farha, Destruction of chemical warfare agents using metal–organic frameworks, Nature Materials 14 (5) (2015) 512–516. doi:10.1038/nmat4238.

[20] S.-Y. Moon, Y. Liu, J. T. Hupp, O. K. Farha, Instantaneous hydrolysis of nerve-agent simulants with a six-connected zirconium-based metal–organic framework, Angewandte Chemie International Edition 54 (23) (2015) 6795–6799. doi:10.1002/anie.201502155.

[21] S. Webb, F. Coulon, T. Temple, A critical review of liquid, low toxicity chemical warfare agent simulants: Enhancing accuracy, safety, and methodological approaches for sampling, Journal of Hazardous Materials 492 (2025) 138021. doi:10.1016/j.jhazmat.2025.138021.

[22] M. Agrawal, D. F. Sava Gallis, J. A. Greathouse, D. S. Sholl, How useful are common simulants of chemical warfare agents at predicting adsorption behavior?, The Journal of Physical Chemistry C 122 (45) (2018) 26061–26069. doi:10.1021/acs.jpcc.8b08856.

[23] J. Lavoie, S. Srinivasan, R. Nagarajan, Using cheminformatics to find simulants for chemical warfare agents, Journal of Hazardous Materials 194 (2011) 85–91. doi:10.1016/j.jhazmat.2011.07.077.

[24] H. S. Chi, E. Kim, J. Kim, K.-D. Byun, K. Kang, Chemical protective suit performance against TICs and CWAs: permeation resistance and PPE safety evaluation, Next Research 2 (3) (2025) 100688. doi:10.1016/j.nexres.2025.100688.

[25] T. James, S. Wyke, T. Marczylo, S. Collins, T. Gaulton, K. Foxall, R. Amlôt, R. Duarte-Davidson, Chemical warfare agent simulants for human volunteer trials of emergency decontamination: A systematic review, Journal of Applied Toxicology 38 (1) (2018) 113–121. doi:10.1002/jat.3527.

[26] T. James, S. Collins, T. Marczylo, Identification of novel simulants for toxic industrial chemicals and chemical warfare agents for human decontamination studies: A systematic review and categorisation of physicochemical characteristics, International Journal of Environmental Research and Public Health 18 (16) (2021) 8681. doi:10.3390/ijerph18168681.

[27] S. Grimme, C. Bannwarth, P. Shushkov, A robust and accurate tight-binding quantum chemical method for structures, vibrational frequencies, and noncovalent interactions of large molecular systems parametrized for all spd-block elements (z = 1–86), Journal of Chemical Theory and Computation 13 (5) (2017) 1989–2009. doi:10.1021/acs.jctc.7b00118.

[28] C. Bannwarth, S. Ehlert, S. Grimme, GFN2-xTB—an accurate and broadly parametrized self-consistent tight-binding quantum chemical method with multipole electrostatics and density-dependent dispersion contributions, Journal of Chemical Theory and Computation 15 (3) (2019) 1652–1671. doi:10.1021/acs.jctc.8b01176.

[29] C. Bannwarth, E. Caldeweyher, S. Ehlert, A. Hansen, P. Pracht, J. Seibert, S. Spicher, S. Grimme, Extended tight-binding quantum chemistry methods, WIREs Computational Molecular Science 11 (2) (2020) e1493. doi:10.1002/wcms.1493.

[30] A. Watson, D. Opresko, R. Young, V. Hauschild, Development and application of acute exposure guideline levels (AEGLs) for chemical warfare nerve and sulfur mustard agents, Journal of Toxicology and Environmental Health, Part B 9 (3) (2006) 173–263. doi:10.1080/15287390500194441.

[31] K. Kang, D.-H. Lee, J. Kim, J. Yoo, Condition-dependent risk cartography of brominated disinfection by-products in seawater desalination, Journal of Environmental Chemical Engineering 14 (5) (2026) 125179. doi:10.1016/j.jece.2026.125179.

[32] F. Urbina, F. Lentzos, C. Invernizzi, S. Ekins, Dual use of artificial-intelligence-powered drug discovery, Nature Machine Intelligence 4 (3) (2022) 189–191. doi:10.1038/s42256-022-00465-9.

[33] D. Kim, J. Yoo, K. Kang, Exposure and disclosure in countering CBRN threats: implications for nonproliferation strategy, The Nonproliferation Review 33 (4-6) (2026) 283–306. doi:10.1080/10736700.2026.2685340.

[34] Y. Jang, J. Kim, D.-H. Lee, J. Yoo, J. Park, K. Kang, Hybrid QSAR modeling of LD50 in organophosphorus nerve agents: a mechanistic approach using DFT and molecular docking, Computational Toxicology 35 (2025) 100363. doi:10.1016/j.comtox.2025.100363.

[35] S. C. Lee, S. Y. Kim, W. S. Lee, S. Y. Jung, B. W. Hwang, D. Ragupathy, D. D. Lee, S. Y. Lee, J. C. Kim, Effects of textural properties on the response of a SnO_2-based gas sensor for the detection of chemical warfare agents, Sensors 11 (7) (2011) 6893–6904. doi:10.3390/s110706893.

[36] A. T. Seitzinger, J. A. Genovese, Chemical agent simulant training composition, U.S. Patent US 7,129,094 B1, assigned to the United States Department of the Army. Filed 16 May 2003; granted 31 October 2006 (2006).
URL https://patents.google.com/patent/US7129094B1/en

[37] D.-H. Lee, S. A. Valenzuela, M. N. Dominguez, M. Otsuka, D. J. Milliron, E. V. Anslyn, A self-degradable hydrogel sensor for a nerve agent tabun surrogate through a self-propagating cascade, Cell Reports Physical Science 2 (9) (2021) 100552. doi:10.1016/j.xcrp.2021.100552.

[38] Y. J. Jang, O. G. Tsay, D. P. Murale, J. A. Jeong, A. Segev, D. G. Churchill, Novel and selective detection of tabun mimics, Chemical Communications 50 (56) (2014) 7531–7534. doi:10.1039/C4CC02689F.

[39] G. Landrum, et al., RDKit: Open-source cheminformatics, https://www.rdkit.org, version 2026.03.6. Cited by name, version and project URL: the Zenodo DOIs registered for RDKit identify individual releases (10.5281/zenodo.10398 is release 2014.03.1) and no version-independent concept DOI could be verified, so none is given here (2026).

[40] S. A. Wildman, G. M. Crippen, Prediction of physicochemical parameters by atomic contributions, Journal of Chemical Information and Computer Sciences 39 (5) (1999) 868–873. doi:10.1021/ci9903071.

[41] P. Ertl, B. Rohde, P. Selzer, Fast calculation of molecular polar surface area as a sum of fragment-based contributions and its application to the prediction of drug transport properties, Journal of Medicinal Chemistry 43 (20) (2000) 3714–3717. doi:10.1021/jm000942e.

[42] L. B. Kier, Shape indexes of orders one and three from molecular graphs, Quantitative Structure-Activity Relationships 5 (1) (1986) 1–7. doi:10.1002/qsar.19860050102.

[43] D. Rogers, M. Hahn, Extended-connectivity fingerprints, Journal of Chemical Information and Modeling 50 (5) (2010) 742–754. doi:10.1021/ci100050t.

[44] S. Kim, J. Chen, T. Cheng, A. Gindulyte, J. He, S. He, Q. Li, B. A. Shoemaker, P. A. Thiessen, B. Yu, L. Zaslavsky, J. Zhang, E. E. Bolton, PubChem 2025 update, Nucleic Acids Research 53 (D1) (2024) D1516–D1525. doi:10.1093/nar/gkae1059.

[45] A. J. Williams, C. M. Grulke, J. Edwards, A. D. McEachran, K. Mansouri, N. C. Baker, G. Patlewicz, I. Shah, J. F. Wambaugh, R. S. Judson, A. M. Richard, The CompTox chemistry dashboard: a community data resource for environmental chemistry, Journal of Cheminformatics 9 (1) (2017) 61. doi:10.1186/s13321-017-0247-6.

[46] S. Ehlert, M. Stahn, S. Spicher, S. Grimme, Robust and efficient implicit solvation model for fast semiempirical methods, Journal of Chemical Theory and Computation 17 (7) (2021) 4250–4261. doi:10.1021/acs.jctc.1c00471.

[47] S. Wang, J. Witek, G. A. Landrum, S. Riniker, Improving conformer generation for small rings and macrocycles based on distance geometry and experimental torsional-angle preferences, Journal of Chemical Information and Modeling 60 (4) (2020) 2044–2058. doi:10.1021/acs.jcim.0c00025.

[48] S. Riniker, G. A. Landrum, Better informed distance geometry: Using what we know to improve conformation generation, Journal of Chemical Information and Modeling 55 (12) (2015) 2562–2574. doi:10.1021/acs.jcim.5b00654.

[49] T. A. Halgren, Merck molecular force field. I. basis, form, scope, parameterization, and performance of MMFF94, Journal of Computational Chemistry 17 (5-6) (1996) 490–519. doi:10.1002/(sici)1096-987x(199604)17:5/6<490::aid-jcc1>3.0.co;2-p.

[50] A. K. Rappe, C. J. Casewit, K. S. Colwell, W. A. Goddard, W. M. Skiff, UFF, a full periodic table force field for molecular mechanics and molecular dynamics simulations, Journal of the American Chemical Society 114 (25) (1992) 10024–10035. doi:10.1021/ja00051a040.

[51] P. Pracht, F. Bohle, S. Grimme, Automated exploration of the low-energy chemical space with fast quantum chemical methods, Physical Chemistry Chemical Physics 22 (14) (2020) 7169–7192. doi:10.1039/c9cp06869d.

[52] R. G. Parr, L. v. Szentpály, S. Liu, Electrophilicity index, Journal of the American Chemical Society 121 (9) (1999) 1922–1924. doi:10.1021/ja983494x.

[53] K. Wiberg, Application of the Pople–Santry–Segal CNDO method to the cyclopropylcarbinyl and cyclobutyl cation and to bicyclobutane, Tetrahedron 24 (3) (1968) 1083–1096. doi:10.1016/0040-4020(68)88057-3.

[54] R. G. Parr, W. Yang, Density functional approach to the frontier-electron theory of chemical reactivity, Journal of the American Chemical Society 106 (14) (1984) 4049–4050. doi:10.1021/ja00326a036.

[55] W. Yang, W. J. Mortier, The use of global and local molecular parameters for the analysis of the gas-phase basicity of amines, Journal of the American Chemical Society 108 (19) (1986) 5708–5711. doi:10.1021/ja00279a008.

[56] P. Geerlings, F. D. Proft, W. Langenaeker, Conceptual density functional theory, Chemical Reviews 103 (5) (2003) 1793–1874. doi:10.1021/cr990029p.

[57] R. BENIGNI, C. BOSSA, Structure alerts for carcinogenicity, and the *Salmonella* assay system: A novel insight through the chemical relational databases technology, Mutation Research/Reviews in Mutation Research 659 (3) (2008) 248–261. doi:10.1016/j.mrrev.2008.05.003.

[58] S. J. Enoch, C. M. Ellison, T. W. Schultz, M. T. D. Cronin, A review of the electrophilic reaction chemistry involved in covalent protein binding relevant to toxicity, Critical Reviews in Toxicology 41 (9) (2011) 783–802. doi:10.3109/10408444.2011.598141.

[59] C. R. Harris, K. J. Millman, S. J. v. d. Walt, R. Gommers, P. Virtanen, D. Cournapeau, E. Wieser, J. Taylor, S. Berg, N. J. Smith, R. Kern, M. Picus, S. Hoyer, M. H. v. Kerkwijk, M. Brett, A. Haldane, J. F. d. Río, M. Wiebe, P. Peterson, P. Gérard-Marchant, K. Sheppard, T. Reddy, W. Weckesser, H. Abbasi, C. Gohlke, T. E. Oliphant, Array programming with numpy, Nature 585 (7825) (2020) 357–362. doi:10.1038/s41586-020-2649-2.

[60] P. Virtanen, R. Gommers, T. E. Oliphant, M. Haberland, T. Reddy, D. Cournapeau, E. Burovski, P. Peterson, W. Weckesser, J. Bright, S. J. v. d. Walt, M. Brett, J. Wilson, K. J. Millman, N. Mayorov, A. R. J. Nelson, E. Jones, R. Kern, E. Larson, C. J. Carey, İ. Polat, Y. Feng, E. W. Moore, J. VanderPlas, D. Laxalde, J. Perktold, R. Cimrman, I. Henriksen, E. A. Quintero, C. R. Harris, A. M. Archibald, A. H. Ribeiro, F. Pedregosa, P. v. Mulbregt, A. Vijaykumar, A. P. Bardelli, A. Rothberg, A. Hilboll, A. Kloeckner, A. Scopatz, A. Lee, A. Rokem, C. N. Woods, C. Fulton, C. Masson, C. Häggström, C. Fitzgerald, D. A. Nicholson, D. R. Hagen, D. V. Pasechnik, E. Olivetti, E. Martin, E. Wieser, F. Silva, F. Lenders, F. Wilhelm, G. Young, G. A. Price, G.-L. Ingold, G. E. Allen, G. R. Lee, H. Audren, I. Probst, J. P. Dietrich, J. Silterra, J. T. Webber, J. Slavič, J. Nothman, J. Buchner, J. Kulick, J. L. Schönberger, J. V. d. M. Cardoso, J. Reimer, J. Harrington, J. L. C. Rodríguez, J. Nunez-Iglesias, J. Kuczynski, K. Tritz, M. Thoma, M. Newville, M. Kümmerer, M. Bolingbroke, M. Tartre, M. Pak, N. J. Smith, N. Nowaczyk, N. Shebanov, O. Pavlyk, P. A. Brodtkorb, P. Lee, R. T. McGibbon, R. Feldbauer, S. Lewis, S. Tygier, S. Sievert, S. Vigna, S. Peterson, S. More, T. Pudlik, T. Oshima, T. J. Pingel, T. P. Robitaille, T. Spura, T. R. Jones, T. Cera, T. Leslie, T. Zito, T. Krauss, U. Upadhyay, Y. O. Halchenko, Y. Vázquez-Baeza, Scipy 1.0:

fundamental algorithms for scientific computing in python, Nature Methods 17 (3) (2020) 261–272. doi:10.1038/s41592-019-0686-2.

[61] W. McKinney, Data structures for statistical computing in python, 2010, pp. 56–61. doi:10.25080/majora-92bf1922-00a.

[62] J. D. Hunter, Matplotlib: A 2d graphics environment, Computing in Science & Engineering 9 (3) (2007) 90–95. doi:10.1109/mcse.2007.55.

[63] National Institute for Occupational Safety and Health, Chemical warfare agent (CWA) simulant program for respirator materials resistance testing, Tech. rep., National Personal Protective Technology Laboratory, NIOSH, dated 31 July 2002 (Jul. 2002).
URL https://archive.cdc.gov/www_cdc_gov/niosh/npptl/standardsdev/cbrn/apr/concepts/pdfs/simulantjul31-508.pdf

[64] D. S. Lawrence, K. A. Van Houten, Simulants for chemical and explosive threats, Johns Hopkins APL Technical Digest 37 (4) (2025).

[65] D.-H. Lee, D.-N. Lee, J.-I. Hong, A fluorescent probe for a lewisite simulant, New Journal of Chemistry 40 (11) (2016) 9021–9024. doi:10.1039/C6NJ01658H.